

30. Ch 15 Support Vector Machines (SVM)

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Support Vector Machines (SVM)

Mostly following ch 15 in Shalev-Schwartz & Ben-David

SVM use linear classifiers,

well-understood theory and good practical performance

- First analyzed in the simpler separable case
by Vapnik & Chervonenkis 1964

- Margin-based analysis & popularization & name due to

Corinna Cortes & Vladimir Vapnik 1995
now head of google research

and Boser, Guyon, Vapnik '92

Can be used w/ kernels ... to be discussed next chapter

Setup: $S = (z_i)_{i=1}^m$, $z_i = (\vec{x}_i, y_i)$, $y_i \in \{\pm 1\}$
binary classification

Assuming data is (linearly) separable (for now)

$h(x) = \langle w, x \rangle + b$ is our model for $\mathcal{H}$ (aka L_d in ch. 9)

Assume $\exists (w, b)$ s.t. $y_i = \underset{\substack{\uparrow \\ \text{in ch. 9}}}{\text{sign}}(\langle w, x_i \rangle + b) \forall i \in [m]$
i.e. $y_i(\langle w, x_i \rangle + b) > 0$

As before, often work w/ "homogeneous" case ($b=0$)

by including a new dimension, $w \leftarrow \begin{bmatrix} w \\ b \end{bmatrix} \overset{\uparrow}{\underset{\downarrow}{d}} \quad x \leftarrow \begin{bmatrix} x \\ 1 \end{bmatrix}$

In that case, in ch 9 we derived the ERM problem
for half space classifiers as the linear program (LP)

Find w s.t. $y_i \langle w, x_i \rangle \geq 1 \forall i \in [m]$

(or... $\min_w 0^T w$ s.t. $y_i \langle w, x_i \rangle \geq 1 \forall i \in [m]$)

and also $\text{vcdim}(\mathcal{H} = \text{sign} \circ L_d) = \underline{d+1}$

(p. 2) SVM

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Goal of SVM: - remove dependence on ambient dimension d } esp. important for kernel methods
- extend to non-separable case

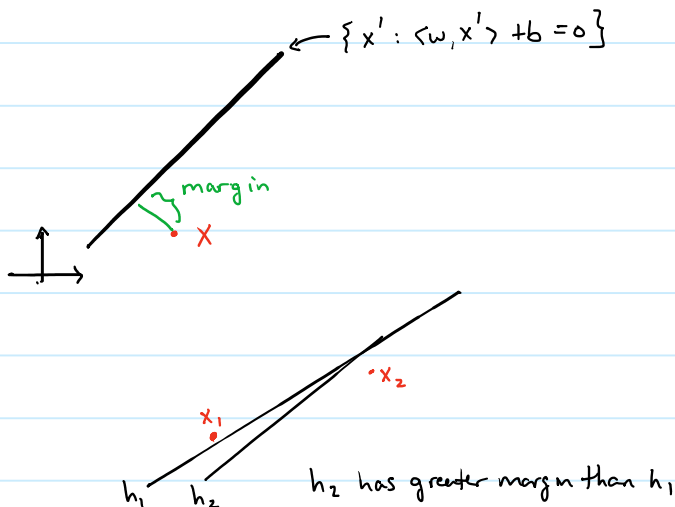
... related issue:

in separable case, which w to choose (since ERM not unique)

Main idea: maximize the margin

Def the (geometric) margin between a point x and the hyperplane defined by $\{x' : \langle w, x' \rangle + b = 0\}$ is the distance* between them, and the margin between $\{x_i\}_{i \in [m]}$ and the hyperplane is the minimum of the individual margins

* in the appropriate Hilbert space norm



Fact (Claim 15.1) Let $h = \{v : \langle \hat{w}, v \rangle + b = 0\}$, then if $\|\hat{w}\| = 1$, $\text{dist}(x, h) = |\langle \hat{w}, x \rangle + b|$

Proof $\text{dist}^2(x, h) = \min \|x - v\|^2$ st $\langle \hat{w}, v \rangle + b = 0$

use Lagrange multipliers:

$$\mathcal{L}(v, \lambda) = \|x - v\|^2 + \lambda (\langle \hat{w}, v \rangle + b)$$

$$\min_v \mathcal{L}(v, \lambda) \Rightarrow 0 = 2(v - x) + \lambda \hat{w}$$

$$\text{so } v = x - \frac{\lambda}{2} \hat{w}, \text{ find } \lambda \text{ by plugging into equality}$$

$$\langle \hat{w}, x - \frac{\lambda}{2} \hat{w} \rangle + b = 0$$

$$\Rightarrow \langle \hat{w}, x \rangle - \frac{\lambda}{2} + b = 0, \quad \frac{\lambda}{2} = \langle \hat{w}, x \rangle + b$$

$$\text{so } v = x - (\langle \hat{w}, x \rangle + b) \hat{w}$$

$$\text{thus } \|x - v\| = |\langle \hat{w}, x \rangle + b| \cdot \|\hat{w}\| = 1 \quad \square$$

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Thus, if we want the maximum margin linear classifier,

solve

$$\max_{(\hat{w}, b) \in \mathbb{R}^{d+1}} \quad \min_{i \in [m]} |\langle \hat{w}, x_i \rangle + b| \quad \text{margin}$$

not good $\downarrow$

s.t. $y_i (\langle \hat{w}, x_i \rangle + b) > 0$ correct classification

$\|\hat{w}\| = 1$ wlog

can merge (Exercise 15.1)

$$\max_{(\hat{w}, b)} \quad \min_{i \in [m]} y_i (\langle \hat{w}, x_i \rangle + b)$$

$y_i \in \{\pm 1\} \Rightarrow \pm 0.1$

$\|\hat{w}\| = 1$

or, even nicer,

due to strict (since strictly) convexity, min is unique

HARD-SVM $(w_0, b_0) = \arg \min \|w\|^2 \text{ s.t. } y_i (\langle w, x_i \rangle + b) \geq 1$

Convex quadratic

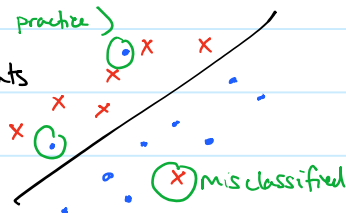
and $\tilde{w} \leftarrow w_0 / \|w_0\|$ and $\tilde{b} \leftarrow b_0 / \|w_0\|$

(theoretical ... not used in practice)

You can analyze via Rademacher complexity
but we're going to proceed to "Soft"-SVM

No longer assuming data is separable: Soft-SVM
(the version you use in practice)

Allow some misclassified data points



SOFT-SVM

$$\min_{w, b, \xi} \lambda \|w\|^2 + \frac{1}{m} \sum \xi_i$$

slack variable $\downarrow$

s.t. $(\forall i \in [m]) \quad y_i (\langle w, x_i \rangle + b) \geq 1 - \xi_i$

$\xi_i > 0$ means (x_i, y_i) misclassified

so penalize $\frac{1}{m} \sum \xi_i$ in objective

NO: $\xi_i \geq 1$ means misclassified $(y_i \in \{\pm 1\})$

and parameter $\lambda > 0$ describes how much we're willing
to tolerate misclassifications (since all else being equal,
small $\|w\|$ is good: smaller R.C.)

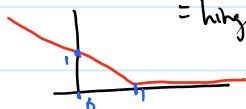
Can rewrite Soft-SVM in 2 ways

(i) in terms of sample risk of a loss function

$$\text{hinge}((w, b), (x, y)) = \max(0, 1 - y(\langle w, x \rangle + b))$$

$$= \text{hinge}(y(\langle w, x \rangle + b))$$

$\hat{y}$



(we've seen before:
a convex surrogate
for 0-1 loss)

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$$\text{and } \hat{L}_S^{\text{hinge}}(w, b) = \frac{1}{m} \sum_{i=1}^m \text{hinge}(w, b, (x_i, y_i))$$

then can write Soft-SVM as

$$\min_{w, b} \underbrace{\hat{L}_S^{\text{hinge}}(w, b)}_{\text{ERM}} + \underbrace{\lambda \|w\|^2}_{\text{RLM}}$$

proof of equivalence ... straight-forward:

$$\text{constraint } y_i (\langle w, x_i \rangle + b) \geq 1 - \xi_i$$

is really an equality $\square$

② via duality

(SOFT-)
SVM
DUAL

$$\max_{\alpha \in \mathbb{R}^m, \alpha \geq 0} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

or in matrix notation

$$\mathbb{1}^T \alpha - \frac{1}{2} \alpha^T G \alpha,$$

$$G = \begin{bmatrix} \text{diag}(y) & \\ & \text{diag}(y) \end{bmatrix} \quad G \succeq 0 \quad \text{So convex}$$

x_m row i is x_i^T

Comments

- In optimization, sometimes primal or dual is easier to solve. Some solvers solve both simultaneously (usually at the end you can convert a dual sol'n into a primal sol'n)

$$\text{in our case, } w = \sum_{i=1}^m \alpha_i y_i x_i \quad (\text{see also Thm 15.8})$$

most $\alpha_i = 0 \dots$

the x_i corresponding to $\alpha_i > 0$
are "Support vectors"

- The dual problem only involves $\bar{x}_i$ in that it needs $\langle x_i, x_j \rangle \forall i, j \in [m]$ and nothing more.

This is the basis of the **kernel trick** that we'll discuss in the next chapter

- There are many algorithms to solve Soft SVM since it's a nicely structured convex problem
 - Book shows a SGD (subgradient) algo.
 - libLinear is well-known (partly for its examples)
 - Scikit Learn, etc., have implementations

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Soft-SVM Sample complexity

Fact: L^{hinge} , for $b=0$, (ie, $f(w) = [1 - y \langle w, x \rangle]_+$)

is $\|x\|$ -Lipschitz

$$\text{(proof } \delta f(w) = \begin{cases} -y \cdot \tilde{x} & \text{if } \dots \\ 0 & \text{else } \dots \end{cases} \text{ either way, } |y|=1, \text{ so } \|\delta f(w)\| \leq \|x\|)$$

Thus, letting $b=0$ wlog,
we can apply the results from last chapter on
RLM, using our reformulation of Soft-SVM as

$$\min_w \lambda \|w\|^2 + \hat{L}_S^{\text{hinge}}(w)$$

Corollary 15.7 Let D be a distribution over $X \times Y$ with $Y = \{0, 1\}$ ~~$Y = \{\pm 1\}$~~
and assume $\|x\| \leq \rho \forall x \in X$. Then if $A(S)$ is the soft-SVM algo.,

$$\textcircled{1} \forall u \in \mathbb{R}^d, \mathbb{E}_{S \sim D^m} L_D^{\text{hinge}}(A(S)) \leq L_D^{\text{hinge}}(u) + \lambda \|u\|^2 + \frac{2\rho^2}{\lambda m}$$

$$\textcircled{2} \forall u \in \mathbb{R}^d, \mathbb{E}_{S \sim D^m} L_D^{0-1}(A(S)) \leq L_D^{\text{hinge}}(u) + \lambda \|u\|^2 + \frac{2\rho^2}{\lambda m}$$

(follows immediately since $L^{\text{hinge}} \geq L^{0-1}$)

$$\textcircled{3} \forall B > 0, \text{ choosing } \lambda = \sqrt{\frac{2\rho^2}{B^2 m}} \text{ then}$$

$$\mathbb{E}_S L_D^{0-1}(A(S)) \leq \mathbb{E}_S L_D^{\text{hinge}}(A(S)) \leq \min_{\|u\| \leq B} L_D^{\text{hinge}}(u) + \sqrt{\frac{8\rho^2 B^2}{m}}$$

Proof $\textcircled{1}$ via Corollary 13.8, $\textcircled{3}$ easy or via Cor. 13.9

Discussion:

Good result if $\min_{\|u\| \leq B} L_D^{\text{hinge}}(u)$ is small

Sometimes true (and may improve on VC-dim. based analysis)